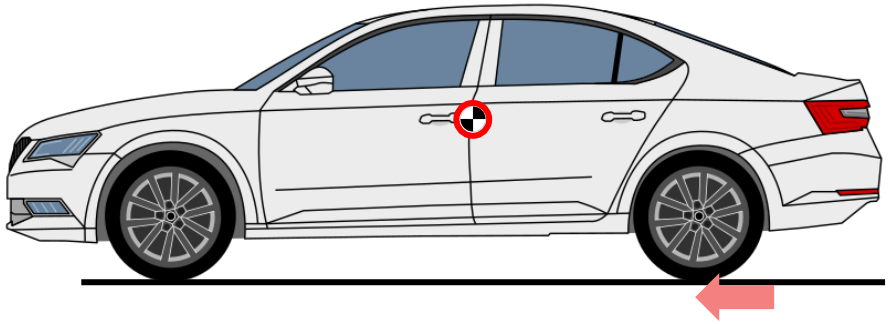
Acceleration Problem

An Rear wheel drive vehicle with a mass 2500 kg is just starting motion
And it must attain an acceleration of 0 -50 kmph in 5 seconds
What would be the total tractive force required at the rear wheels to achieve it ?
Considering negligible Rolling resistance.



Acceleration in m/s^2

$$\frac{v-u}{t} = a$$

$$v = 50\ kmph = 13.889\ ms^{-1}$$

$$t = 5\ s \qquad u = 0$$

$$\frac{13.889 - 0}{5} = 2.778\ ms^{-2}$$

Inertia to be overcome to achieve acceleration

$$Tractive\ force\ - Rolling\ resistance\ force = mass\ * acceleration$$

$$Tractive\ force\ = mass\ * acceleration$$

$$Max\ Tractive\ force\ = 6945\ N$$